	SVKM's NMIMS **Mukesh Patel School of Technology Management & Engineering /** **School of Technology Management & Engineering**	
B. Tech CE	**Lab and Workbook**	**Academic Year- 2026-27**
Year:-Third	**Subject:-** Drone Technology	**Semester: Fifth**

Experiment: 1

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Aim: To understand the various components of a quadcopter, their functions, and how they work together to achieve stable flight.

Prerequisite: Basic knowledge of UAVs & their applications, familiarity with drone communication protocols

Learning Outcome:

After completing the lab, students will be able to:

1. **Identify and explain the function of each quadcopter component.**
2. **Understand how different parts work together for stable flight.**
3. **Analyze the role of the flight controller**
4. **Verify correct motor rotation and propeller orientation.**

Theory:

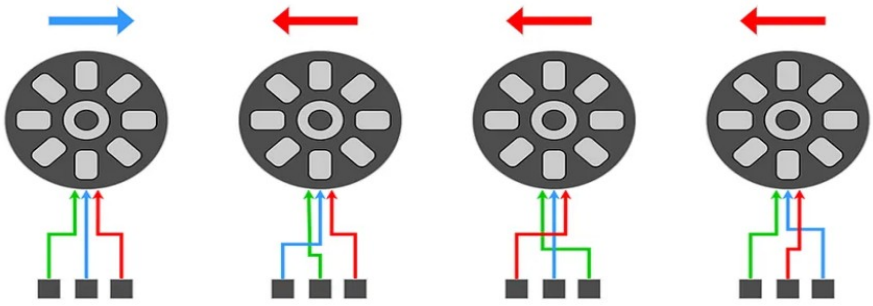
A quadcopter is an unmanned aerial vehicle (UAV) that uses four rotors for lift and maneuverability. The quadcopter's stability and control depend on its components, which include:

Frame

- The structural backbone that holds all components together.
- Made from materials like carbon fiber, aluminum, or plastic.

Motors

- Brushless DC motors (BLDC) provide thrust and control drone movement.
- Each motor's rotation direction (clockwise or counterclockwise) is critical for stability.



Electronic Speed Controllers (ESCs)

- Regulate power supply to motors based on flight controller signals.
- Convert throttle input into appropriate motor speed.

Flight Controller

- The brain of the quadcopter, processes sensor data and controls motor outputs.
- Examples: PX4, Betaflight, Ardupilot.

Propellers

- Generate thrust and lift; must be correctly paired (clockwise and counterclockwise).
- Made of plastic or carbon fiber.

Battery and Power Distribution Board (PDB)

- LiPo (Lithium Polymer) batteries provide power to motors and electronics.
- The PDB distributes power from the battery to different components.

Radio Transmitter and Receiver

- The transmitter (remote controller) sends commands, and the receiver interprets them.
- Works on protocols like SBUS, PPM, or DSM.

Sensors (IMU, GPS, etc.)

- **IMU (Inertial Measurement Unit):** Includes accelerometer and gyroscope for stability.
- **GPS Module:** Provides location data for autonomous navigation.

PART B

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Prog/Yr/Sem: B.Tech CE / Third Year / Fifth Sem	**Batch:** C1
Date of Experiment: 15/10/24	**Date of Submission:** 15/10/24

Task to be performed

Task 1: Component Identification & Explanation

Task 2: Understanding System Integration

Task 3: Flight Controller Analysis

Task 4: Motor Rotation & Propeller Orientation Verification

Task 1: Component Identification & Explanation

Component	Function	Typical specification / example
Frame	Structural body that holds motors, battery, FC and payload; arms carry motors	X / H / + type; carbon fibre / aluminium / plastic (e.g. F450: 450 mm wheelbase)
BLDC motors (×4)	Produce thrust; speed variation gives control	2212 / 920–1000 KV, 3-phase, 2 CW + 2 CCW
ESC (×4)	Convert FC's PWM signal into 3-phase power for BLDC motor	20–30 A, 2–4S, BEC optional
Flight Controller	Brain: reads sensors, runs stabilisation & navigation, drives ESCs	Pixhawk (ArduPilot / PX4), Betaflight boards

Propellers	Convert motor torque into lift; paired CW/CCW	10×4.5 plastic or carbon fibre
LiPo battery + PDB	Battery supplies energy; PDB distributes it to ESCs and electronics	3S/4S 11.1–14.8 V, 3000–5000 mAh, 20C+
TX / RX	Pilot commands sent wirelessly and decoded	2.4 GHz, 6+ channels, SBUS / PPM / DSM
Sensors (IMU, GPS, baro, compass)	IMU = attitude & rotation; GPS = position; baro = altitude; compass = heading	MPU6000 IMU, u-blox GPS, MS5611 barometer

Task 2: Understanding System Integration

Signal and power flow of a quadcopter:

```
Pilot > Transmitter ))) 2.4 GHz ((( Receiver
Receiver --(SBUS/PPM)--> Flight Controller
Sensors (IMU, GPS, compass, baro) --> Flight Controller
Flight Controller --(PWM/DShot)--> 4 x ESC --> 4 x BLDC motor
BLDC motor --> Propeller --> Thrust and lift
LiPo battery --> PDB --> ESCs (high current)
LiPo battery --> Power module / BEC (5 V) --> FC, RX, GPS
```

- The pilot's stick input is transmitted to the receiver and passed to the flight controller as roll, pitch, throttle and yaw commands.
- The IMU continuously measures the real angular rate and acceleration; the flight controller compares them with the commanded values using PID control.
- The FC computes a separate throttle value for each motor and sends it to the ESCs; the ESCs drive the motors at that speed.
- Stable flight is maintained by this closed loop that runs hundreds of times per second. GPS, barometer and compass add position, altitude and heading hold.

Motion by varying motor speed (Quad-X):

Movement	Motor speed change
Hover / climb / descend	All four motors increase / equal / decrease together
Roll (left / right)	Increase speed on one side's motors, decrease on the other side
Pitch (forward / backward)	Increase rear (front) motors, decrease front (rear) motors
Yaw	Speed up the CW pair and slow the CCW pair (or vice-versa) – unbalances reaction torque

Task 3: Flight Controller Analysis

- The flight controller (e.g. Pixhawk running ArduPilot) has a 32-bit processor, an IMU (3-axis gyroscope + accelerometer), barometer and compass; external GPS is added through a serial port.
- Inputs: RC receiver (SBUS/PPM), sensors, battery monitor (power module), telemetry from ground station. Outputs: PWM / DShot signals on MAIN OUT 1–4 to the ESCs, plus telemetry and logs.
- Control: sensor fusion (EKF) estimates attitude and position; cascaded PID loops (angle → rate → motor mixer) calculate motor outputs.
- Safety: pre-arm checks, arming switch, failsafes for radio loss, low battery and GPS loss (RTL / Land).

- Examples compared: ArduPilot and PX4 (autonomous missions, GPS, Pixhawk hardware) and Betaflight (fast acro control, racing quads).

Task 4: Motor Rotation & Propeller Orientation Verification

In a Quad-X, diagonal motors rotate in the same direction. Looking from the top, with front of the drone at the top (ArduCopter motor order):

```
      FRONT
3 (CW)     1 (CCW)
      \   /
       [FC]
      /   \
2 (CCW)    4 (CW)
      REAR
```

Motor	Position	Rotation	Propeller
1	Front-right	CCW	CCW (reverse “R”) prop
2	Rear-left	CCW	CCW (reverse “R”) prop
3	Front-left	CW	CW (normal) prop
4	Rear-right	CW	CW (normal) prop

Verification procedure (props removed):

- Remove all propellers and power the drone from a battery.
- Use Mission Planner → Initial Setup → Optional Hardware → Motor Test; run each motor at a low throttle (≈ 5–10 %) for a few seconds.
- Observe each motor’s rotation direction against the table above. If a motor spins the wrong way, swap any two of its three ESC–motor wires.
- Fit the propellers: CCW prop on CCW motors and CW prop on CW motors with the flat/curved leading edge facing the direction of rotation and the writing (size) facing up; tighten the nut.

Important: *never test with propellers fitted while the drone is on a bench.*

Observations and Learning:

- The quadcopter works as a system: frame → motors/ESCs → FC → sensors/RC → battery; every part depends on the others.
- Opposite-direction (CW/CCW) motor pairs cancel torque, so the drone yaws only when we deliberately change the speed of one pair.
- The flight controller is the brain: it combines IMU data and pilot commands using PID loops to keep the drone stable.
- ESCs convert the weak FC signal into the high-current 3-phase power required by the motors.
- Motor direction must be tested before fitting propellers, and propeller type must match the motor.
- The power module/BEC provides stable 5 V power to the FC and receiver independent of motor load.

Conclusion:

The components of a quadcopter – frame, BLDC motors, ESCs, flight controller, propellers, LiPo battery with PDB, transmitter/receiver and sensors – were identified and their functions explained. It was understood how they work together through a closed control loop to give stable flight, the role of the flight controller was analysed, and the motor rotation direction and propeller orientation were verified (diagonal motors same direction). All four learning outcomes were attained.